[16] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 53728–53741. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/a85b405ed65c6477a4fe8302b5e06ce7-Paper-Conference.pdf.

[17] Xiangyu Qi, Kaixuan Huang, Ashwinee Panda, Peter Henderson, Mengdi Wang, and Prateek Mittal. Visual adversarial examples jailbreak aligned large language models, 2023.

[18] Nicholas Carlini, Milad Nasr, Christopher A Choquette-Choo, Matthew Jagielski, Irena Gao, Anas Awadalla, Pang Wei Koh, Daphne Ippolito, Katherine Lee, Florian Tramer, et al. Are aligned neural networks adversarially aligned? *arXiv preprint arXiv:2306.15447*, 2023.

[19] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.

[20] Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2023.

[21] Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. Jailbreaking leading safety-aligned llms with simple adaptive attacks, 2024.

[22] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to! *arXiv preprint arXiv:2310.03693*, 2023.

[23] Qiusi Zhan, Richard Fang, Rohan Bindu, Akul Gupta, Tatsunori Hashimoto, and Daniel Kang. Removing rlhf protections in gpt-4 via fine-tuning. *arXiv preprint arXiv:2311.05553*, 2023.

[24] Yangsibo Huang, Samyak Gupta, Mengzhou Xia, Kai Li, and Danqi Chen. Catastrophic jailbreak of open-source llms via exploiting generation. *arXiv preprint arXiv:2310.06987*, 2023.

[25] Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base LLMs: Rethinking alignment via in-context learning. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=wxJ0eXwwda.

[26] Xiao Zhang and Ji Wu. Dissecting learning and forgetting in language model finetuning. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=tmsqb6WpLz.

[27] Xuandong Zhao, Xianjun Yang, Tianyu Pang, Chao Du, Lei Li, Yu-Xiang Wang, and William Yang Wang. Weak-to-strong jailbreaking on large language models. *arXiv preprint arXiv:2401.17256*, 2024.

[28] Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, LILI YU, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. Lima: Less is more for alignment. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 55006–55021. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/ac662d74829e4407ce1d126477f4a03a-Paper-Conference.pdf.

[29] Andy Zou, Zifan Wang, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.

[30] Haize Labs. A trivial jailbreak against llama 3. https://github.com/haizelabs/llama3-jailbreak, 2024.

[31] Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.

[32] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Hex-phi: Human-extended policy-oriented harmful instruction benchmark. `https://huggingface.co/datasets/LLM-Tuning-Safety/HEx-PHI`, 2023.

[33] Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization, 2022.

[34] Anthropic. Prefill Claude's response. `https://docs.anthropic.com/claude/docs/prefill-claudes-response`, 2024.

[35] Luxi He, Mengzhou Xia, and Peter Henderson. What's in your" safe" data?: Identifying benign data that breaks safety. *arXiv preprint arXiv:2404.01099*, 2024.

[36] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. `https://github.com/tatsu-lab/stanford_alpaca`, 2023.

[37] Xuechen Li, Tianyi Zhang, Yann Dubois, Rohan Taori, Ishaan Gulrajani, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Alpacaeval: An automatic evaluator of instruction-following models. `https://github.com/tatsu-lab/alpaca_eval`, 2023.

[38] Andrew Peng, Michael Wu, John Allard, Logan Kilpatrick, and Steven Heidel. Gpt-3.5 turbo fine-tuning and api updates, 8 2023. URL `https://openai.com/blog/gpt-3-5-turbo-fine-tuning-and-api-updates`.

[39] Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. Kto: Model alignment as prospect theoretic optimization. *arXiv preprint arXiv:2402.01306*, 2024.

[40] Sidharth Mudgal, Jong Lee, Harish Ganapathy, YaGuang Li, Tao Wang, Yanping Huang, Zhifeng Chen, Heng-Tze Cheng, Michael Collins, Trevor Strohman, Jilin Chen, Alex Beutel, and Ahmad Beirami. Controlled decoding from language models. In *International Conference on Machine Learning (ICML)*, 2024.

[41] Charles Dugas, Yoshua Bengio, François B'elisle, Claude Nadeau, and Ren'e Garcia. Incorporating second-order functional knowledge for better option pricing. In *Proceedings of the 13th International Conference on Neural Information Processing Systems (NIPS'00)*, pages 451–457. MIT Press, 2000. Since the sigmoid $h$ has a positive first derivative, its primitive, which we call softplus, is convex.

[42] Bogdan Gliwa, Iwona Mochol, Maciej Biesek, and Aleksander Wawer. SAMSum corpus: A human-annotated dialogue dataset for abstractive summarization. In *Proceedings of the 2nd Workshop on New Frontiers in Summarization*, pages 70–79, Hong Kong, China, November 2019. Association for Computational Linguistics. doi:10.18653/v1/D19-5409. URL `https://www.aclweb.org/anthology/D19-5409`.

[43] b mc2. sql-create-context dataset, 2023. URL `https://huggingface.co/datasets/b-mc2/sql-create-context`. This dataset was created by modifying data from the following sources: [59, 60].

[44] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[45] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[46] Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms. *arXiv preprint arXiv:2401.06373*, 2024.

[47] Xianjun Yang, Xiao Wang, Qi Zhang, Linda Petzold, William Yang Wang, Xun Zhao, and Dahua Lin. Shadow alignment: The ease of subverting safely-aligned language models. *arXiv preprint arXiv:2310.02949*, 2023.

[48] Pranav Gade, Simon Lermen, Charlie Rogers-Smith, and Jeffrey Ladish. Badllama: cheaply removing safety fine-tuning from llama 2-chat 13b. *arXiv preprint arXiv:2311.00117*, 2023.

[49] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, et al. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.

[50] Andy Zou. Breaking llama guard. `https://github.com/andyzoujm/breaking-llama-guard`, 2023.

[51] Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. Assessing the brittleness of safety alignment via pruning and low-rank modifications. *arXiv preprint arXiv:2402.05162*, 2024.

[52] Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P Dick, Hidenori Tanaka, Edward Grefenstette, Tim Rocktäschel, and David Scott Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. *arXiv preprint arXiv:2311.12786*, 2023.

[53] Zhangchen Xu, Fengqing Jiang, Luyao Niu, Jinyuan Jia, Bill Yuchen Lin, and Radha Poovendran. Safedecoding: Defending against jailbreak attacks via safety-aware decoding. *arXiv preprint arXiv:2402.08983*, 2024.

[54] Alekh Agarwal, Sham M Kakade, Jason D Lee, and Gaurav Mahajan. On the theory of policy gradient methods: Optimality, approximation, and distribution shift. *Journal of Machine Learning Research*, 22(98):1–76, 2021.

[55] Brijen Thananjeyan, Ashwin Balakrishna, Suraj Nair, Michael Luo, Krishnan Srinivasan, Minho Hwang, Joseph E Gonzalez, Julian Ibarz, Chelsea Finn, and Ken Goldberg. Recovery rl: Safe reinforcement learning with learned recovery zones. *IEEE Robotics and Automation Letters*, 6 (3):4915–4922, 2021.

[56] Robert R Burridge, Alfred A Rizzi, and Daniel E Koditschek. Sequential composition of dynamically dexterous robot behaviors. *The International Journal of Robotics Research*, 18(6): 534–555, 1999.

[57] Aaron D Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *2019 18th European control conference (ECC)*, pages 3420–3431. IEEE, 2019.

[58] Peter Henderson, Eric Mitchell, Christopher Manning, Dan Jurafsky, and Chelsea Finn. Self-destructing models: Increasing the costs of harmful dual uses of foundation models. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*, pages 287–296, 2023.

[59] Victor Zhong, Caiming Xiong, and Richard Socher. Seq2sql: Generating structured queries from natural language using reinforcement learning. *CoRR*, abs/1709.00103, 2017.

[60] Tao Yu, Rui Zhang, Kai Yang, Michihiro Yasunaga, Dongxu Wang, Zifan Li, James Ma, Irene Li, Qingning Yao, Shanelle Roman, et al. Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-sql task. *arXiv preprint arXiv:1809.08887*, 2018.

[61] Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*, 2022.

[62] Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

# A Detailed Setups of Our Experiments

## A.1 Compute Resources

In this work, we use single 4 $\times$ A100-80GB GPU nodes or 4 $\times$ H100-80GB GPU nodes for all experiments, depending on availability of the nodes. On each node, our experiments use up to 8 CPU cores and 256GB memory, but overall the experiments are not CPU intensive tasks.

## A.2 General Configurations

**Decoding Parameters.** Throughout this paper, we use the top-p sampling with a temperate of 0.9 and a top-p parameter of 0.6 by default for decoding outputs from LLMs in our experiments. The only case where we do not follow this default configuration is the decoding parameters exploit experiment where the exploit itself needs to take use of different parameters by its design [24].

**Safety Evaluation.** As also mentioned in Section 2.1, we use the GPT-4 based judge to evaluate the safety of model outputs, following the setup of Qi et al. [22]. Specifically, in such an evaluation pipeline, we pass (input, output) pairs to the GPT-4-Turbo model, and prompt the model to evaluate the harmfulness level of the output. The model will output a score ranging from 1 to 5, with higher score indicating being more harmful. When reporting ASR in our experiments, we report the ratio of outputs that get the highest score 5 (identical to the harmfulness rate metric in Qi et al. [22]). By default, HEx-PHI safety benchmark [32] is used for safety evaluation. The only exception is the experiments in Table 2, where we add additional evaluation on AdvBench for GCG attack evaluation [29] and MaliciousInstruct for decoding parameters exploit [24]. These two additional safety evaluation datasets are used in the original papers of the two work, and we report results on both HEx-PHI and these additional safety evaluation datasets for a more complete reference.

## A.3 Details of Data Augmentation Experiments

Here, we describe the implementation details of the data augmentation experiments in Section 3.1.

**Safety Data.** As noted in Eqn 2, we use a safety dataset $D_H$ to keep generating safety recovery examples. To construct it, we first collect 256 harmful instructions. These instructions are mostly collected from the red-teaming data provided by Ganguli et al. [61]. We make sure they do not overlap with any of the safety evaluation datasets that we used in this paper, i.e., HEx-PHI [32], AdvBench [29], and MaliciousInstruct [24]. Then, we generate refusal answers for each harmful instruction using the initial Llama-2-7B-Chat model. We also collect the corresponding harmful responses for these instructions using a jailbroken version of the model (jailbroken through fine-tuning attacks per Qi et al. [22]). This results in the dataset $D_H$ with 256 examples of triplets $(\boldsymbol{x}, \boldsymbol{h}, \boldsymbol{r})$.

**Utility Data.** To prevent the decrease of model utility during the data augmentation fine-tuning, we also take benign instructions from the Alpaca [36] dataset. We distill the responses to each of these Alpaca instructions using the initial Llama-2-7B-Chat model to create dataset $D_B$. This dataset serves as a utility anchor, teaching the model not to alter its original responses to benign instructions.

**Training details with Eqn 2.** In the implementation of the augmentation fine-tuning as per Eqn 2, we set the number of prefilled tokens $k$ to follow a distribution $\mathcal{P}_k$, where $k = 0$ with a $50\%$ probability, and $k$ is uniformly sampled from $[1, 100]$ with a $50\%$ probability. We set $\alpha = 0.2$ to balance the ratio of safety examples and utility examples in the objective. In batch-wise training, this is implemented by randomly sampling 16 examples from $D_H$ and 64 examples from $D_B$ in each batch. Using this objective, we train the model for 10 epochs on $D_H$ with a learning rate of $2 \times 10^{-5}$ using the AdamW optimizer (with the default configurations of the optimizer).

**AlpacaEval.** We also evaluate the utility of the augmented fine-tuned model with AlpacaEval [37], which is reported as a winrate against the text-davinci-003 model (the default reference baseline for AlpacaEval). Specifically, we use the 1.0 version of AlpacaEval without length control. To ensure the setup is consistent with the safety evaluation, the official system prompt (with a focus on safety) of Llama-2-7B-Chat is used when running AlpacaEval. We note that the win rates here can therefore be generally lower than the numbers in official Alpaca leaderboard in which the safety system prompt is not applied. Under this evaluation, we note that the augmented fine-tuned model achieves a winrate of $49.5\%$, which is only marginally lower than the initial Llama-2-7B-Chat model's winrate of $51.8\%$.